CS19541 – COMPUTER NETWORKS LAB

PRACTICAL – 13

IMPLEMENT YOUR OWN PING PROGRAM

Aim

To create a basic ping program that checks whether a host is reachable and calculates its round-trip response time.

Algorithm

1. Get a host name or IP address from the user.
2. Convert the host name into its corresponding IP address.
3. Open a raw socket that uses the ICMP protocol.
4. Build and transmit an ICMP Echo Request packet.
5. Wait for an Echo Reply and calculate the round-trip time.
6. Print the response time or a timeout message, repeat if needed, and close the socket.

Python Code

```
import socket
import struct
import time

host = input("Enter host name or IP address: ")
ip = socket.gethostbyname(host)
sock = socket.socket(socket.AF_INET, socket.SOCK_RAW, socket.IPPROTO_ICMP)
sock.settimeout(2)

for i in range(4):
    packet_id = i
    packet = struct.pack("!BBHHH", 8, 0, 0, packet_id, 1)
    checksum = 0
    for j in range(0, len(packet), 2):
        checksum += (packet[j] << 8) + packet[j + 1]
    checksum = (checksum >> 16) + (checksum & 0xffff)
    checksum = ~checksum & 0xffff
    packet = struct.pack("!BBHHH", 8, 0, checksum, packet_id, 1)
    start = time.time()
    sock.sendto(packet, (ip, 0))
    try:
        sock.recvfrom(1024)
        end = time.time()
        print("Reply from", ip, "time =", round((end - start) * 1000, 2), "ms")
    except socket.timeout:
        print("Request timed out")

sock.close()
```

Input

Enter a host name or IP address, for example: 8.8.8.8.

Output

The program prints the reply and round-trip time in milliseconds. If a response is not received before the timeout, it displays “Request timed out”.

Note

Creating raw ICMP sockets may require Administrator or root permissions. Some networks and hosts block ICMP traffic, so a timeout alone does not prove that the destination host is offline.

Result

Thus, a simple ping program was implemented to send ICMP Echo Requests and display the response time.